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Late-Season Nitrogen and Grain Protein in Rainfed Winter Wheat

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Abstract

Late-season nitrogen (N) application is often recommended to raise grain protein in winter wheat (*Triticum aestivum* L.). We tested a late application of 3 kg N per decare (30 kg N/ha) at the flag-leaf stage in addition to a basal dose, in six rainfed environments over three seasons. Late N increased grain protein concentration by 0.8 percentage points on average ($P < 0.01$). Grain yield was not significantly affected by late N in any environment ($P > 0.05$).

Keywords: winter wheat, late nitrogen, grain protein, rainfed

Results and Discussion

Across environments, late N raised grain protein from 11.9% to 12.7%. The effect on grain yield was small and not significant: mean yield was 4.21 t/ha with late N and 4.17 t/ha without it. In the driest environment, late N had no measurable effect on either trait.

These results suggest that late-season N in rainfed systems should be regarded as a quality measure. It should not be expected to increase grain yield.

Conclusion

Late N at flag leaf improved grain protein but did not increase grain yield under rainfed conditions.